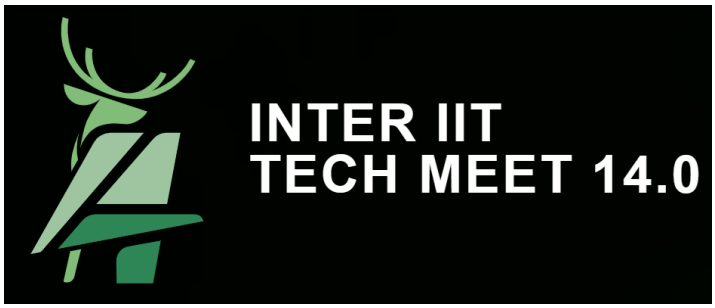


TEAM ID - 42

Delta-Sigma ADC ASIC Development



End Term Report

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1 Introduction

1.1 Overview of Delta-Sigma ADC

A Delta-Sigma Analog-to-Digital Converter (ADC) is a high-resolution data conversion architecture widely used in applications that require high precision at low bandwidths, such as instrumentation, biomedical signal processing, and communication systems [1]. Unlike conventional Nyquist-rate ADCs, Delta-Sigma ADCs operate by oversampling the input signal at a much higher frequency than the Nyquist rate and applying a technique known as noise shaping to push quantization noise out of the signal band [2, 3].

The core of a Delta-Sigma ADC is a feedback-based modulator consisting of an integrator, a coarse quantizer (typically 1-bit or multi-bit), and a digital-to-analog feedback loop [4]. This modulator produces a high-frequency bitstream whose average value represents the input signal. A digital decimation filter is then used to remove out-of-band noise and reduce the data rate, producing a high-resolution digital output.

The main advantages of Delta-Sigma ADCs include excellent linearity, high effective resolution (often exceeding 16 bits), and relaxed requirements for analog anti-aliasing filters. These properties make ADCs highly suitable for VLSI implementation and low-power mixed-signal systems [5].

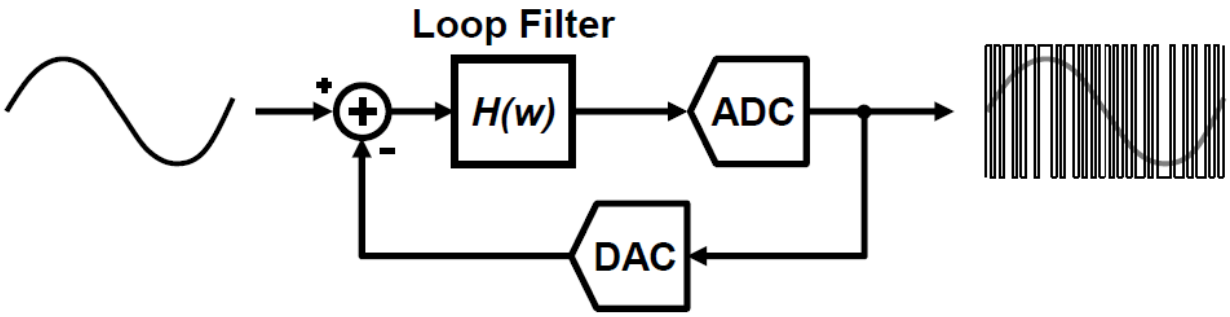


Figure 1: BASIC DELTA SIGMA MODULATOR

1.2 Project Objective

The primary objective of this project is to study, model, and design a high-resolution Delta-Sigma Analog-to-Digital Converter (ADC) system with a focus on achieving high Effective Number of Bits (ENOB) at practical sampling rates suitable for space and precision instrumentation applications. The project aims to explore different Delta-Sigma modulator architectures and digital decimation filter structures to understand their impact on performance, complexity, and hardware feasibility.

Another key objective is to develop a complete digital back-end by designing and implementing the decimation filter at the Register Transfer Level (RTL) using Verilog and validating its functionality through simulation. The project also aims to bridge system-level modeling and hardware realization by comparing MATLAB/Simulink models with fixed-point HDL implementations.

Ultimately, the goal is to establish a design methodology that supports reliable VLSI implementation of Delta-Sigma ADC architectures, leading toward synthesis, layout, and post-layout verification suitable for ASIC/SoC integration.

2 Architecture and Methodology

The design methodology followed a top-down flow beginning with system-level behavioral modeling and progressing toward hardware description and implementation. Three Delta–Sigma modulator architectures Continuous-Time (CT), Discrete-Time (DT), and Hybrid CT–DT were modeled in MATLAB/Simulink using a matched loop

order, oversampling ratio, and quantizer configuration. The bitstream output from each architecture was recorded and used as the input for performance analysis.

Spectral evaluation was performed for each model using the power spectral density (PSD), signal-to-noise ratio (SNR), and qualitative inspection of noise-shaping behavior. These simulations provided insight into the distribution of quantization noise, noise-shaping characteristics, and the spectral properties of the modulator outputs. The behavioral modeling environment allowed rapid parameter exploration and observation of resulting changes in spectral response.

To investigate the influence of non-ideal mechanisms, additional simulations were carried out with finite Opamp gain and low-frequency noise(Flicker Noise). The resulting spectra were analyzed to understand how noise affects in-band performance, noise leakage, and overall spectral behavior under different operating conditions.

A digital decimation filter was then modeled to recover the baseband signal from the modulator output. Multiple filter structures were explored, including SINC, FIR, and IIR configurations, with variations in order and coefficient quantization. The filters were analyzed based on their ability to attenuate out-of-band quantization noise while maintaining passband fidelity and meeting bandwidth constraints.

Following algorithm-level validation, the decimation filter was implemented at the Register Transfer Level (RTL) using hardware description languages such as Verilog. Functional verification was conducted using test benches to compare RTL output with behavioral simulation output, ensuring logical correctness and consistency with the modeled functionality. Fixed-point representation and word-length configurations were incorporated during RTL development to align with hardware constraints.

After functional verification, the RTL description was synthesized to a gate-level netlist using standard cell libraries. Area, timing, and power reports generated during synthesis were examined to assess whether the implementation met performance and resource constraints.

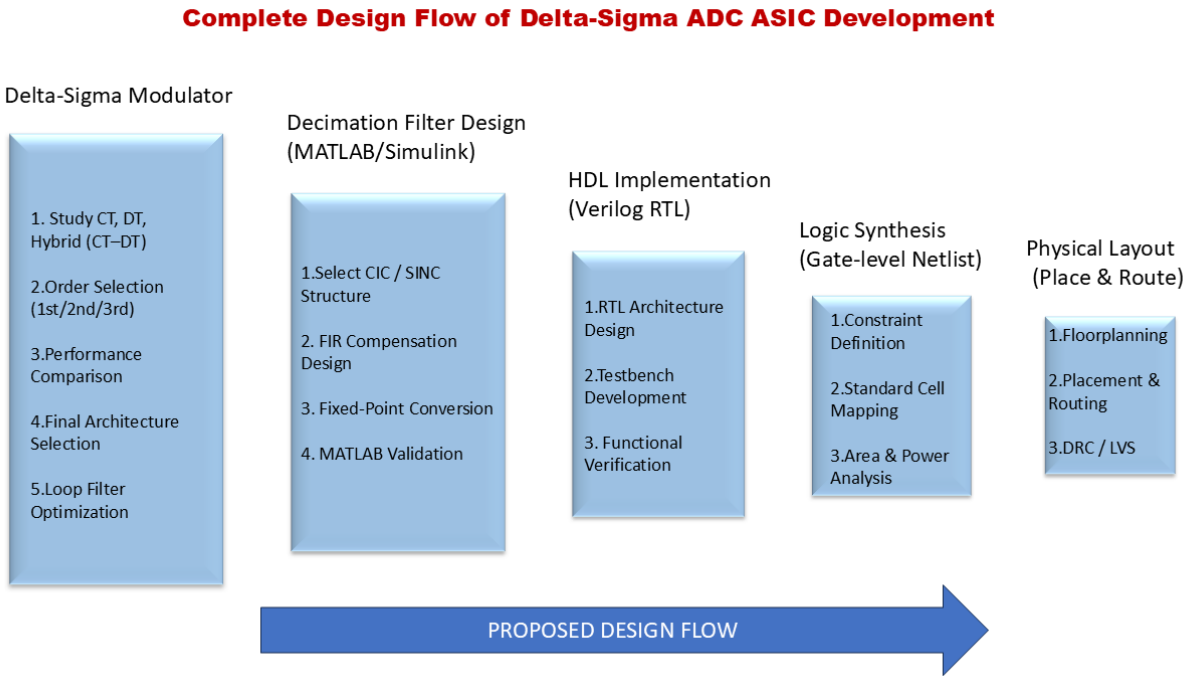


Figure 2: System-level methodology: Behavioral modeling, spectral analysis, noise study, filter design, RTL implementation, synthesis, and layout.

3 Modulator Study

3.1 Modulator Types Explored

In this work, three third-order Delta–Sigma modulator architectures were examined—Discrete-Time (DT), Continuous-Time (CT), and Hybrid CT–DT—using behavioral Simulink models. All architectures were simulated at a sampling frequency of 256 kHz. The input signal bandwidth was limited to 1 kHz, implying a Nyquist rate of 2 kHz. Based on this, the modulator operated at an oversampling ratio (OSR) of 128, defined as $OSR = \frac{f_s}{2BW}$, which provides sufficient noise-shaping headroom for high-resolution operation without incurring excessive simulation complexity.

A third-order loop was selected to provide strong in-band noise shaping while maintaining stable dynamics and manageable implementation complexity. For a single-bit modulator, the theoretical signal-to-noise ratio (SNR) of an L-th order loop is approximated by:

$$SNR \approx 7.78 + (2L+1) \cdot 10\log_{10}(OSR),$$

$$SINAD = 6.02 \cdot ENOB + 1.76$$

and achieving an ENOB greater than 16 bits requires a SINAD of approximately $98\,$dB, obtained from $SINAD = 6.02 \cdot 16 + 1.76$.

Third-order noise shaping provides approximately 18–$20\,$dB SNR improvement per doubling of OSR, enabling practical SNR values on the order of 95–$105\,$dB after accounting for non-ideal effects such as finite gain, noise, and timing uncertainties. This makes the chosen architecture suitable for high-resolution, low-bandwidth sensing applications.

Table 1: Simulation Parameters Used for Behavioral Modeling

Parameter	**Value**
Sampling Frequency (f_s)	256 kHz
Signal Bandwidth (BW)	1 kHz
Nyquist Rate ($2BW$)	2 kHz
Oversampling Ratio (OSR)	128
Modulator Order	3rd Order
Quantizer Resolution	1-bit
Architecture Types	DT, CT, Hybrid CT–DT
Simulation Environment	MATLAB/Simulink
Integrator Type	Behavioral (Discrete/Continuous)
Noise Modeling	Ideal + Non-ideal perturbations
FFT Length	Variable
Performance Metrics	PSD,SNR,Harmonics

The DT architecture was modeled using discrete-time integrator and quantizer blocks, enabling direct manipulation of loop coefficients and observation of noise-transfer-function behavior [4, 5]. The CT architecture was obtained by mapping the discrete-time loop filter to its continuous-time equivalent using pulse-invariant transformation, allowing the study of continuous-time noise shaping, anti-aliasing behavior, and sensitivity to timing-related effects such as jitter and loop delay [6, 7]. To explore intermediate behavior, a hybrid CT–DT modulator was constructed with continuous-time front-end filtering followed by discrete-time loop shaping, based on methodologies proposed in hybrid Sigma–Delta literature [8]. Each architecture was simulated under identical OSR and loop parameters to isolate the impact of architectural differences rather than order-dependent performance.

For each modulator, power spectral density (PSD) plots were generated from the output bitstream using FFT-based estimation to observe quantization noise distribution and spectral shaping trends. The DT model exhibited the expected high-pass noise shaping, whereas the CT architecture demonstrated stronger spectral roll-off due to continuous-time dynamics. The hybrid CT–DT model displayed combined characteristics, with smoother low-frequency behavior and additional spectral variation at higher frequencies resulting from

the interaction between continuous-time filtering and discrete-time quantization. PSD analysis was used to compare in-band noise, harmonic content, and susceptibility to non-ideal effects under a uniform simulation setup, providing qualitative insight into the operational behavior of each architecture.

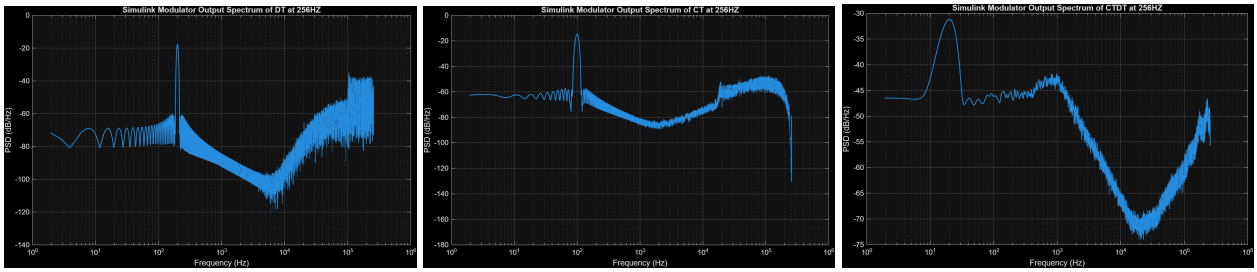


Figure 3: PSD plots of DT, CT, and CT–DT Delta-Sigma Modulators

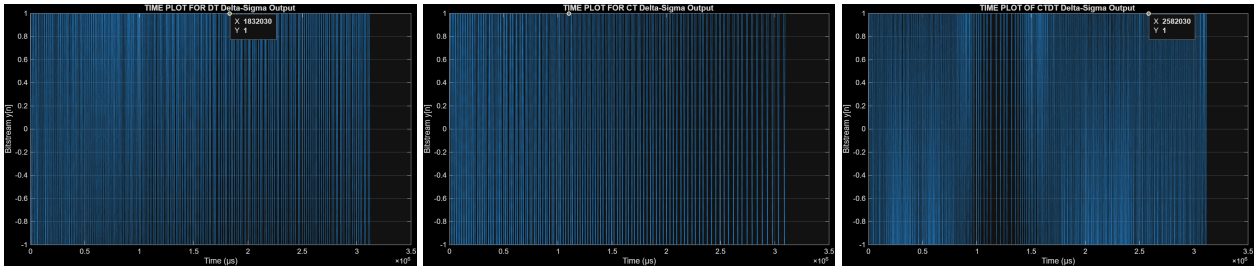


Figure 4: Time-domain bit-stream plots for DT, CT, and CT–DT modulators.

Table 2: Comparison of DT, CT, and CT–DT Delta–Sigma Modulator Architectures

Requirement / Criterion	**DT**	**CT**	**CT–DT**
Low bandwidth operation	✓	–	–
High bandwidth operation	–	✓	✓
High resolution (ENOB)	✓	✓	✓
Low analog design effort	✓	–	–
Low jitter sensitivity	✓	–	✓
Robustness to excess loop delay	✓	–	–
Low power at high bandwidth	–	✓	✓
Ease of modeling / simulation	✓	–	–
Design and verification complexity	Low	High	Very High
Suitability for first prototype	✓	–	–
Maximum achievable performance	✓	–	–

Selected Architecture and Justification

Based on the comparative behavioral analysis, the Discrete-Time (DT) architecture was selected for further development and integration in this work. Although the CT and hybrid CT–DT models demonstrated desirable spectral shaping characteristics, the DT architecture provided a more predictable and controllable design framework under the given oversampling and resolution requirements. In the DT model, loop coefficients are defined directly in the z-domain, allowing deterministic control over the noise-transfer function and straightforward adjustment of stability and dynamic range without requiring continuous-time pulse response modeling or delay compensation [4]. Furthermore, DT modeling demonstrated lower sensitivity to timing uncertainties, such as clock jitter and loop delay, which were observed to degrade spectral performance in the CT and CT–DT simulations. From a methodological standpoint, the DT structure offered simpler parameter tuning, faster simulation convergence, and more consistent correlation between analytical predictions and FFT-based performance metrics such as SNR, ENOB, and harmonic content. These factors made the DT architecture more

suitable for iterative exploration of quantization effects, filter optimization, and eventual hardware mapping, while still meeting the high-resolution objective associated with an OSR of 128.

3.2 ENOB vs Sampling Rate Analysis

To study the impact of output data rate on resolution, the discrete-time third-order modulator was simulated for multiple decimated output sampling rates in the range of 0.5 ksps to 2 ksps, while keeping the modulator clock fixed at 256 kHz (OSR = 128 for a 1 kHz signal bandwidth). For each operating point, the 1-bit modulator bitstream was passed through the corresponding decimation filter to generate a multi-bit baseband output, and the Effective Number of Bits (ENOB) was evaluated at the filter output using a custom MATLAB script, `enob_calculator.m`. The script performs coherent FFT-based spectral analysis on the steady-state output sequence, computes the Signal-to-Noise and Distortion Ratio (SINAD), and converts it to ENOB using

$$ENOB = \frac{\text{SINAD} - 1.76}{6.02}.$$

This post-processing flow was kept identical for all configurations to ensure consistent evaluation and enable a fair comparison of ENOB as a function of output sampling rate. Table 3 summarizes the ENOB values obtained for each output sampling rate. ENOB increases significantly with increasing output sampling frequency, indicating that higher data rates retain a larger portion of the shaped spectral content within the band of interest. At low output rates (0.5–0.7 ksps), aggressive decimation removes substantial in-band information and exposes residual quantization noise, resulting in very low ENOB. In contrast, higher output rates (1.5–2.0 ksps) preserve more in-band energy and improve SINAD, yielding substantially higher resolution.

Table 3: ENOB vs Output Sampling Frequency

Output Sampling Frequency (ksps)	**ENOB (bits)**
2.0	21.03
1.5	20.38
1.2	18.69
1.0	18.54
0.7	11.82
0.5	8.18

Figure 5 illustrates the same trend graphically. ENOB exhibits a strong positive correlation with output sampling frequency, as higher data rates preserve more of the shaped signal energy and reduce the effective impact of quantization noise, while lower rates intensify noise folding during decimation and result in poorer resolution.

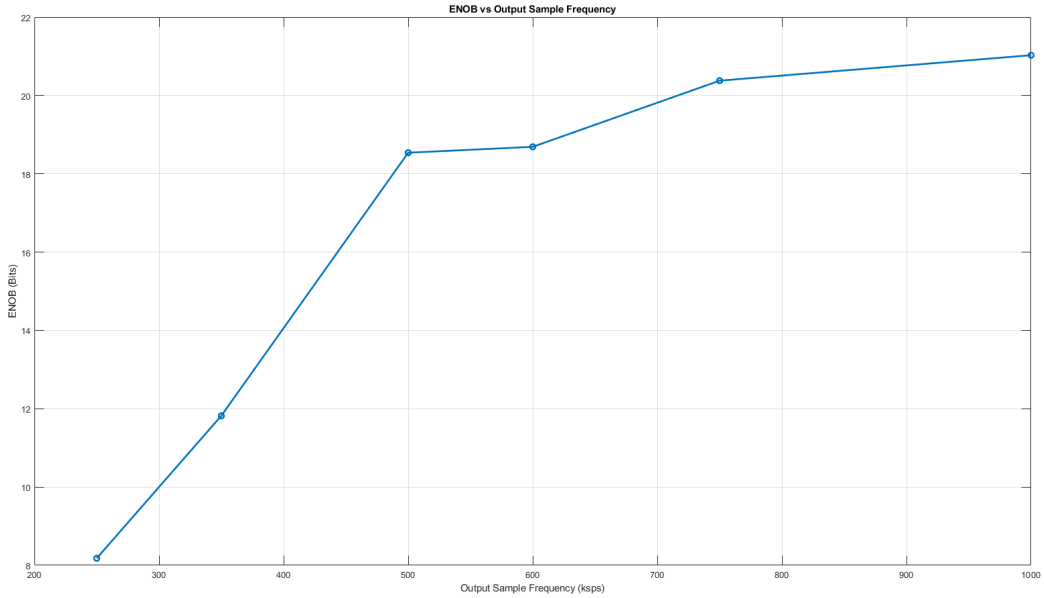


Figure 5: ENOB versus output sampling frequency for the simulated discrete-time third-order modulator. Higher output rates preserve more in-band spectral content, resulting in higher SINAD and improved ENOB.

3.3 Noise Analysis and Mitigation

3.3.1 Flicker Noise Impact

Real-world $\Delta\Sigma$ modulators suffer from non-idealities that degrade in-band performance, especially below 2 kHz. Two dominant sources were modelled:

Finite Op-Amp Gain: Practical integrators exhibit finite DC gain, modelled as:

$$H_{\text{real}}(z) = \frac{1}{z - 0.998}$$

corresponding to ≈54 dB gain. This weakens the NTF zero at DC, increasing quantization noise leakage into the baseband.

Flicker Noise ($1/f$): Pink noise was injected at the modulator input to represent low-frequency CMOS noise. Spectral analysis confirmed loop stability under noise injection and highlighted elevated noise floor at low frequencies.

3.3.2 Noise Mitigation Techniques

Three mitigation methods were examined. **Chopping** shifts noise to higher frequencies but introduces switching artifacts. **Nested chopping** enhances suppression but significantly increases complexity and power. **Auto-zeroing/CDS** cancels offset/noise by sampling-and-subtraction and is naturally suited to discrete-time systems.

3.3.3 Selected Strategy

A compact **multi-stage digital mitigation pipeline** was adopted:

1. **Discrete-time differentiation $(1 - z^{-1})$:** Implements CDS to suppress DC and low-frequency $1/f$ noise.
2. **ZOH spectral shaping:** Provides $\text{sinc}^2(f)$ attenuation of high-frequency aliases prior to decimation.
3. **2nd-order IIR bandpass filter (80–990 Hz):** Final rejection of residual low-frequency noise not suppressed by earlier stages.

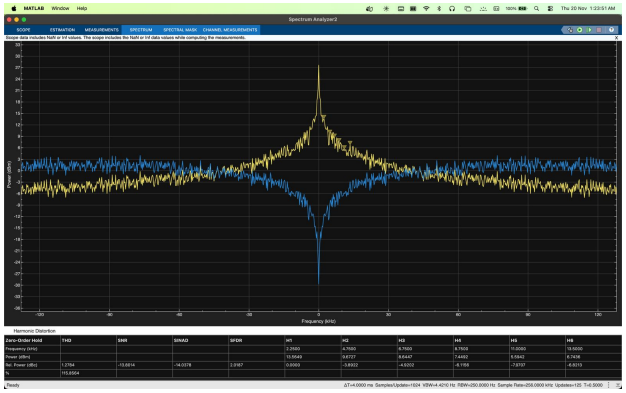


Figure 6: Noise shaping under $1/f$ injection.

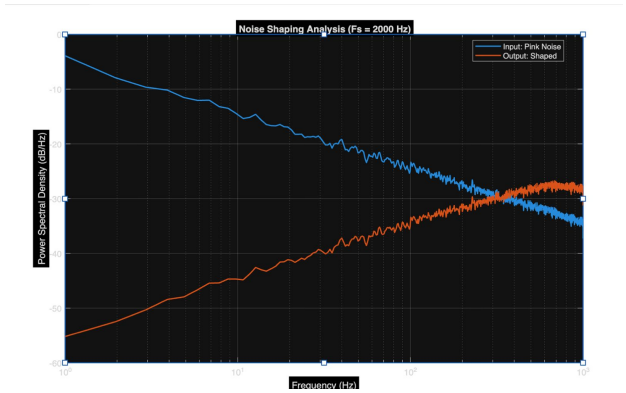


Figure 7: ZOH-induced spectral shaping.

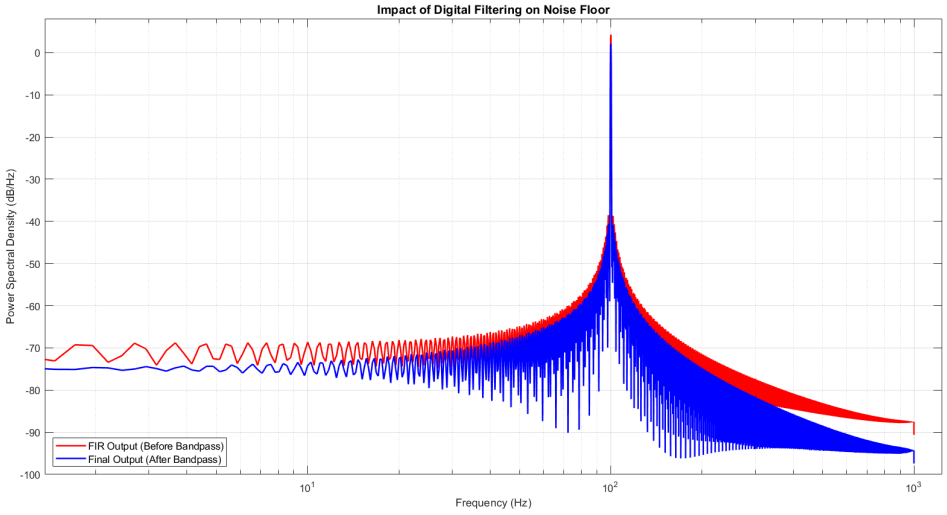


Figure 8: Effectiveness of the proposed mitigation pipeline.

4 Digital / Decimation Filter

Two decimation architectures were developed to process the oversampled 1-bit $\Delta\Sigma$ modulator output. Both begin with a CIC decimator that performs coarse rate reduction but introduces sinc^N passband droop, requiring additional FIR processing to shape the baseband spectrum. The first architecture (Chain 1) follows the conventional SINC-based structure reported in prior work [9], consisting of CIC decimation, FIR compensation, and a halfband FIR stage. While this approach partially corrects passband distortion and provides linear-phase decimation, it exhibits residual ripple, limited stopband attenuation, and higher in-band noise leakage, ultimately constraining ENOB. In contrast, the proposed architecture (Chain 2) employs a CIC decimator followed by a halfband FIR and a bandpass FIR filter. The halfband filter improves passband flatness, while the bandpass stage suppresses low-frequency drift and $1/f$ noise, resulting in lower in-band noise floor, stronger alias rejection, and improved ENOB and SINAD. Overall, the alternative FIR-based chain provides superior spectral shaping and noise suppression compared to the conventional SINC structure.

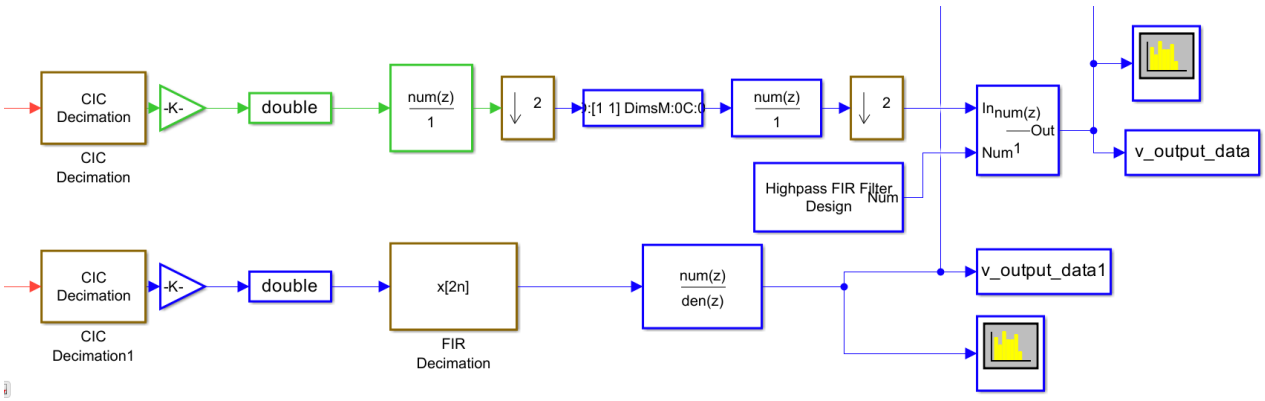


Figure 9: Block diagram of the two decimation filter architectures.

Table 4: Key design parameters of the two decimation filter chains.

Chain 1: SINC-Based Decimation	**Chain 2: Proposed FIR-Based Decimation**
CIC Decimation Filter	
Decimation factor (R): 32 Number of sections (N): 4 Differential delay (M): 1	Decimation factor (R): 64 Number of sections (N): 4 Differential delay (M): 1
FIR Compensation Filter	**FIR Halfband Decimator**
Length: 128 taps Decimation: Yes(factor 2)	Length: 51 taps Decimation: Yes (factor 2)
FIR Halfband Filter (Chain 1)	**FIR Bandpass Filter (Chain 2)**
Length: 61 taps Decimation: No	Order: 2 (biquad) Decimation: No
High-Pass FIR Filter	—
Length: 128 taps Cutoff frequency: 1 Hz	— —

4.1 Comparison of SINC and Proposed FIR-Based Decimation Filters

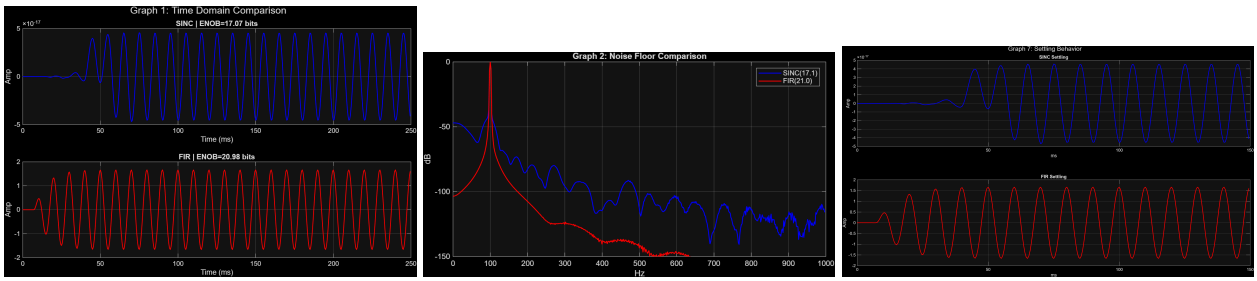


Figure 10: Time-domain, noise-floor, and settling behaviour comparison between SINC and FIR-based decimation filters.

Performance Summary

Filter	SINAD (dB)	ENOB	PHD (deg)	THD (dB)
SINC filter	104.52	17.07	226.236	-81.96
FIR-based filter	136.96	22.46	40.291	-115.40

Table 5: Performance comparison of SINC and FIR-based decimation filters.

A comparative evaluation of the two decimation architectures was carried out using time-domain behavior, noise-floor characteristics, and standard performance metrics. The baseline SINC chain exhibited slower settling, amplitude distortion, and higher residual noise, resulting in an effective resolution of around 17 bits. In contrast, the proposed FIR chain produced a cleaner reconstructed waveform with faster transient response and a significantly lower noise floor (over 30 dB reduction), leading to improved SINAD, reduced harmonic and phase distortion, and an ENOB of approximately 22.5 bits. Coefficient quantization studies further indicated that precision below 10 bits severely degraded performance, while 12–14 bit precision closely matched floating-point behavior; thus, FIR stages were implemented using 16-bit fixed-point coefficients to balance accuracy and hardware cost. Overall, the FIR-based architecture demonstrated superior spectral and temporal characteristics compared to the conventional SINC approach, making it more suitable for high-resolution sensing within the 0–1 kHz band.

4.2 Quantization of Coefficients

A coefficient quantization study was performed to evaluate the impact of fixed-point precision on filter accuracy and overall system resolution. Floating-point FIR coefficients were quantized to word-lengths ranging from 4 to 24 bits, and the resulting outputs were analyzed using the same ENOB estimation flow applied to the decimated modulator data. The results indicate a strong dependence of performance on coefficient precision, particularly in the low-resolution region. For word-lengths below approximately 10 bits, the system exhibits severe degradation in ENOB due to quantization-induced shifts in pole/zero locations and loss of stopband attenuation. Performance improves rapidly between 10 and 14 bits, after which the improvement becomes marginal, as quantization noise falls below the modulator's intrinsic noise floor. The peak ENOB of approximately 23.4 bits occurs at 14-bit precision, with little gain beyond this point.

Metric	4	6	8	10	12	14	16	24
ENOB (bits)	0.00	1.51	10.92	20.13	22.03	**23.37**	22.53	22.44

Table 6: ENOB values for different coefficient bit-widths.

Based on these results, a 16-bit fixed-point representation was selected as a practical compromise between hardware cost and numerical accuracy, providing near-optimal ENOB while avoiding the computational overhead of higher precision arithmetic.

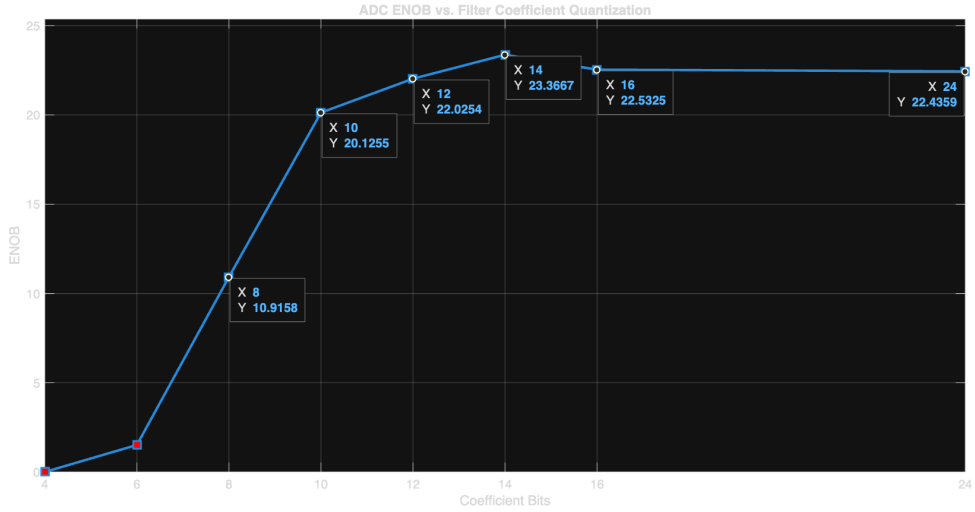


Figure 11: Plot ADC ENOB vs Coefficient Quantization

5 Comparison of Modulator Architectures with Designed Digital Filters

Configuration	SINAD (dB)	ENOB (bits)
CT–DT	100.69	16.43
CT Only	98.11	16.00
DT	**128.39**	**21.03**

Table 7: Comparison of Modulator Architectures with Designed Digital Filter

The results clearly show that the DT-domain filtering achieves significantly superior performance compared to both CT and CT–DT configurations.

6 HDL Design and Implementation

The digital decimation filter chain was implemented at the Register Transfer Level (RTL) using Verilog HDL. The design architecture adheres to the specifications of the Delta-Sigma ADC system, supporting a maximum oversampling frequency (f_s) of 10 MHz and providing high-precision Pulse Code Modulation (PCM) output. The architecture employs a multi-stage decimation topology with a total decimation factor of $D = 128$ to down-sample the single-bit modulated bitstream while maximizing the Signal-to-Noise Ratio (SNR).

6.1 RTL Architecture Overview

The datapath comprises three cascaded processing stages balancing logic depth and memory utilization:

1. **Stage 1: CIC Decimator.** A 4th-order Cascaded Integrator-Comb (CIC) filter performing bulk decimation by a factor of 64.

2. **Stage 2: FIR Decimator.** A 51-tap symmetric Finite Impulse Response (FIR) filter providing an additional decimation factor of 2 (total $D = 128$) and compensating for the passband droop inherent to the CIC transfer function.

3. **Stage 3: IIR Bandpass Filter.** A 2nd-order Infinite Impulse Response (IIR) filter configured to eliminate DC offsets ($z = 1$) and suppress high-frequency noise components.

6.2 Detailed Module Descriptions

6.2.1 CIC Decimator

The CIC module implements a recursive Hogenauer structure comprising four integrator stages operating at the oversampling rate (f_s) and four comb stages operating at the decimated rate ($f_s/64$). A critical design consideration in CIC filters is register overflow due to accumulation. To address this, the internal register width was analytically determined using the Hogenauer bit-growth formula:

$$B_{out} = B_{in} + N \log_2 R \tag{1}$$

where N represents the filter order and R denotes the decimation ratio. The internal precision was extended to 25 bits, ensuring no information loss due to wraparound arithmetic during the integration phase. A gain normalization stage employing bitwise right-shift operations is applied at the output to scale the signal amplitude appropriately prior to the FIR stage.

6.2.2 FIR Decimator

The second stage utilizes a polyphase decomposition structure to implement a 51-tap symmetric FIR filter. Exploiting the decimation-by-2 requirement, the architecture computes output samples only on alternate clock cycles, effectively reducing the Multiply-Accumulate (MAC) operation frequency by half, thereby minimizing dynamic power consumption.

- **Precision Handling:** The arithmetic unit employs a 40-bit accumulator to preserve computational precision during MAC operations, with filter coefficients represented in 16-bit Q1.15 fixed-point format.
- **Saturation Logic:** Overflow protection logic is incorporated to clamp the output signal to the maximum signed integer range defined by the output bus width, preventing arithmetic wraparound errors.

6.2.3 IIR Bandpass Filter

The final stage implements a Direct Form I biquadratic (biquad) structure for the 2nd-order bandpass filter. The design employs fixed-point Q2.14 coefficients to establish zeros at DC and Nyquist frequencies.

- **Limit Cycle Mitigation:** Recursive filter structures are susceptible to limit-cycle oscillations induced by quantization effects. To mitigate this phenomenon, the feedback ($y[n-k]$) and feedforward ($x[n-k]$) paths utilize extended 64-bit intermediate registers, maintaining high internal precision.
- **Output Resolution:** The high-precision accumulator output is truncated to the final output width only at the terminal stage, minimizing quantization error propagation throughout the signal chain.

6.3 Simulation and Verification Results

Functional verification was conducted using a file-based simulation methodology. The RTL implementation was simulated in the Xilinx Vivado environment, with stimulus provided by a testbench ingesting bitstream vectors generated from a MATLAB reference model. This approach ensures bit-accurate correspondence between the Verilog implementation and the theoretical floating-point design.

6.3.1 Signal Reconstruction and Spectral Analysis

The RTL output data was exported to MATLAB for post-processing and spectral analysis. Figure 12 presents the time-domain reconstruction of the analog waveform, demonstrating the filter chain's capability to recover the original signal from the 1-bit density-modulated stream with high fidelity. Spectral analysis of the reconstructed signal via Fast Fourier Transform (FFT) yielded an **Effective Number of Bits (ENOB) of 15.86**, confirming the design's high dynamic range performance and validating the theoretical noise-shaping characteristics.

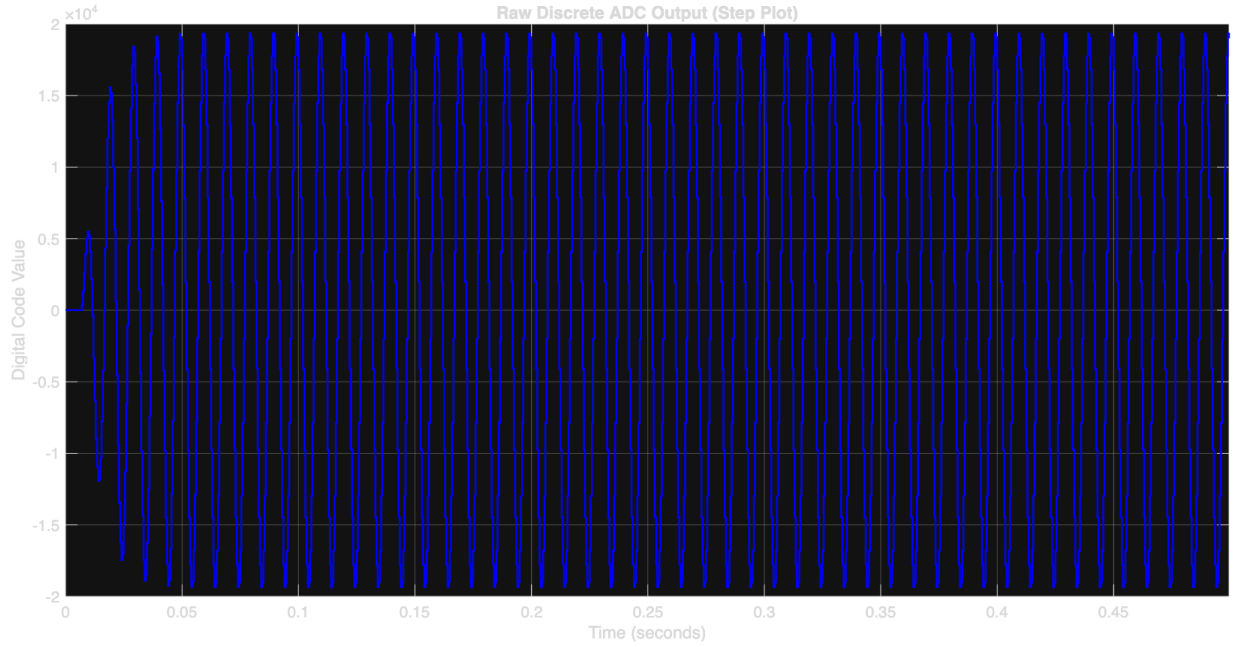


Figure 12: Time-domain reconstruction of RTL output data performed in MATLAB, demonstrating high-fidelity signal recovery from the Delta-Sigma modulated bitstream.

6.4 Timing Feasibility Analysis

Static timing analysis was performed on the synthesized design to verify operation at the target 10 MHz oversampling frequency. The analysis confirms that the logic depth remains well within the required timing budget.

- **Timing Closure:** The design satisfies all setup and hold timing constraints with substantial margin. The analysis yielded a **Worst Negative Slack (WNS) of +86.592 ns**, indicating that the critical path delay is significantly less than the 100 ns clock period (10 MHz operation).
- **Resource Utilization:** The synthesis results indicate efficient hardware utilization, consuming 553 Look-Up Tables (LUTs) and 91 Flip-Flops (FFs), thereby leaving considerable area available for further backend optimization and integration.

7 ASIC Flow

Synthesis Results

Table 8: Synthesis Summary of Key Metrics

Metric	Value	Constraint/Target	Status
Technology	GPDK 0.45μm	N/A	N/A
Total Cell Count	9,520 Gates	N/A	N/A
Clock Period	20.00 ns	20.00 ns	MET
WNS (Worst Negative Slack)	+14.940 ns	≥ 0 ns	PASS
Critical Path Delay	4.030 ns	≤ 19.00 ns	PASS

Area Analysis

The synthesized design occupies approximately **9,520 equivalent gates**. The `firdecimator` module is the largest contributor, accounting for **72.4%** of the total area, which is expected due to the computational cost of multi-tap FIR filtering and MAC operations.

Timing Analysis

The design meets the **50 MHz** clock requirement (20 ns period) with a **WNS of +14.940 ns**, indicating strong timing closure. The **critical path** is **4.030 ns** within the `iirhighpass` module, suggesting potential for higher operating frequencies or relaxed optimization for area/power reduction.

8 Challenges Faced

During the project, we faced several practical and technical difficulties at different stages: behavioural modelling, fixed-point design, RTL coding, and synthesis.

- **Behavioural modelling stability:** Getting a stable Delta–Sigma modulator in Simulink was not straightforward. Small changes in loop coefficients or input amplitude sometimes made the system unstable or caused the output to overflow. Adding non-ideal effects like noise made this worse and required careful tuning and repeated simulations.
- **Fixed-point representation and RTL mapping:** Converting the floating-point MATLAB/Simulink model into fixed-point hardware was challenging. We had to choose word-lengths that were not too small (to avoid large quantization errors and loss of ENOB) and not too large (to avoid extra area and power). Several iterations were needed to fix overflow, scaling, and rounding issues so that the RTL output matched the reference model.
- **Synthesis and timing constraints:** Meeting timing and area goals at the target clock frequency required changes in architecture and coding style. Some arithmetic blocks had to be restructured, and pipeline or resource

9 New Learning

The project enabled several important learnings across system modelling and digital design. At the modelling stage, behavioural simulations provided insight into Delta–Sigma modulation, noise-shaping behaviour, and how oversampling, loop configuration, and non-idealities influence spectral performance. Moving from behavioural models to fixed-point descriptions revealed practical challenges such as word-length selection, coefficient quantization, and handling overflow during arithmetic operations. RTL development and verification offered experience in writing synthesizable Verilog, creating test benches, and comparing results with reference outputs to confirm functional correctness. The synthesis step provided an understanding of how RTL is translated into gates, along with exposure to timing and area estimation, and how early architectural decisions affect resource usage and achievable performance.

10 Limitations & Risks

Even though the system works as intended, there are some limitations and risks that can affect its performance and future scaling.

- **Sensitivity to noise and stability:** Delta–Sigma modulators are very sensitive to small changes in coefficients, extra noise, and delays in the loop. Under extreme conditions, this can reduce in-band performance or even make the modulator unstable.
- **Fixed-point precision limits:** Because the digital filters and RTL design use fixed-point arithmetic, limited word-length can cause rounding and quantization errors. This may reduce the maximum achievable ENOB and increase the chance of overflow or saturation if the signal range is not carefully controlled.

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A Appendix

This appendix provides concise supporting information relevant to the modelling and implementation of the Delta–Sigma ADC presented in this report.

A. Key Definitions

- **Oversampling Ratio (OSR):** Ratio of sampling frequency to twice the signal bandwidth; higher OSR enables improved noise shaping.
- **Noise Shaping:** Technique used in Delta–Sigma modulators to push quantization noise toward high frequencies, enabling low in-band noise.
- **Decimation:** Combination of filtering and downsampling to reduce data rate after oversampling.
- **ENOB:** Effective resolution of an ADC, derived from observed SINAD.
- **SINAD:** Ratio of signal power to total noise and distortion power.

B. Abbreviations

ADC	Analog-to-Digital Converter
CT	Continuous-Time
DT	Discrete-Time
FIR	Finite Impulse Response
IIR	Infinite Impulse Response
CIC	Cascaded Integrator-Comb
PSD	Power Spectral Density
ENOB	Effective Number of Bits
SINAD	Signal-to-Noise and Distortion
RTL	Register Transfer Level

D. Analysis and Verification Method

Behavioural simulations were performed in MATLAB/Simulink with FFT-based measurement of SINAD and ENOB. RTL was verified using a file-driven testbench in Vivado, where modulator bitstreams were applied to the DUT and results were exported back to MATLAB for comparison. Fixed-point arithmetic employed 16-bit coefficients and extended accumulators to minimize quantization error.

E. Limitations and Assumptions

- Analogue non-idealities were modelled behaviourally, not at transistor level.
- Post-layout parasitics and power analysis were not included.